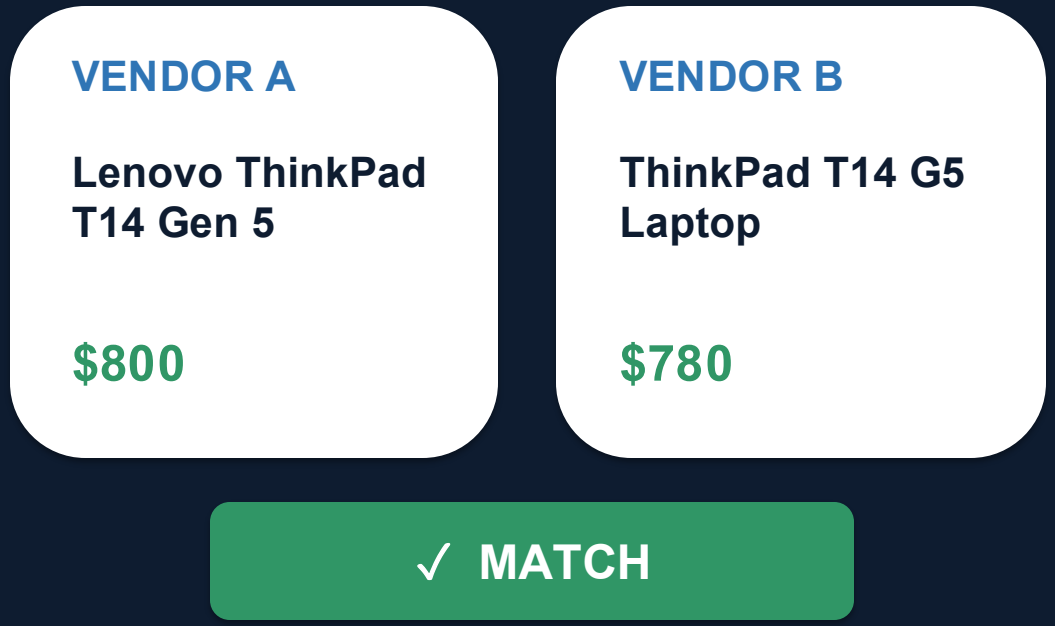
PRODUCT MATCHING

Same product.
Different listing.

Automatically identify matching products across vendors, then make the match useful.



Who cares?

Procurement • distributors • supply chain teams • customers

Goal: reduce manual searching and make cross-vendor comparison faster.

Geana C. Gutierrez Vanegas • Michael A. Maier • Rajesh Yadav • Ugur Sener

RESEARCH

Current methods — and where they fall short

Prior work shows that product text and attributes are strong signals for entity matching.

Köpcke et al. (2012)	Manufacturer, codes, titles & descriptions
Qiu et al. (2015)	Collect and normalize product specifications
Mudgal et al. (2018)	Deep learning for messy or incomplete records
Narayan et al. (2022)	Foundation models for cleaning & matching
Steiner et al. (2025)	Fine-tuned LLMs can improve entity matching

LIMITS TODAY

- Vendor text is inconsistent
- Important fields can be missing
- Traditional models need structured features
- Fine-tuning LLMs adds cost and maintenance

Our opportunity: combine matching with an interactive comparison tool.

INNOVATION

Our approach: make the match useful

We go beyond a yes/no prediction.

1 **Clean vendor data** → 2 **Match products** → 3 **Compare results**

Compare prices

Show which vendor has the lower price.

Fill missing specs

Combine useful information across matched listings.

Show confidence

Let users see how certain each product match is.

Models: start with Logistic Regression and SVM; test a stronger text model if needed.

Why it should work: previous research shows descriptions and product attributes can identify matching items.

How we will know it works

Measure success

Accuracy • Precision • Recall

Compare model performance and whether real matches are identified correctly.

Expected impact

Less manual searching
Faster price/spec comparison
More complete product information

Main risk

Missing attributes and inconsistent vendor naming can reduce matching accuracy.

PAYOFF

Reduce manual effort and make vendor comparison easier.

EXPECTED DIRECT COST

$0 for public data; main cost is team time and compute.

Data source: WDC product matching data; public web data may supplement it if needed.

PROJECT PLAN

11-week plan

Mid-term Checkpoint
Cleaned and integrated dataset ready for analysis.

Final checkpoint
Models + visualization complete

Wk	Task	Owner	Progress	Start	End	21-Sep	28-Sep	5-Oct	12-Oct	19-Oct	26-Oct	2-Nov	9-Nov	16-Nov	23-Nov	30-Nov	7-Dec
1	Finalize Project idea and finding of Data	All	100	9/21/2026	9/28/2026	X	X										
1	Dataset feasibility / ground truth check	All	100	9/21/2026	10/2/2026	X	X										
1	Find and assign 12+ literature papers	All	100	9/21/2026	10/2/2026	X	X										
1	Read and summarize assigned papers	All	100	9/25/2026	10/7/2026	X	X	X									
2	Define algorithm/computation innovation	All	100	9/28/2026	10/7/2026		X	X									
2	Answer all 9 Heilmeier questions	All	100	9/30/2026	10/8/2026		X	X									
2	Define evaluation plan and metrics	All	100	9/30/2026	10/8/2026		X	X									
2	Write proposal + Gantt + Presentation-Internal Deadline	All	100	10/2/2026	10/9/2026		X	X									
3	Proposal revisions / submission buffer	All	100	10/7/2026	10/16/2026			X	X								
4	Clean full dataset	Geana, Rajesh	0	10/14/2026	10/19/2026				X	X							
4	Build interactive visualization prototype	Michael	0	10/14/2026	10/27/2026				X	X	X						
5	Develop proposed algorithm- Use of models LG and SVM	All	0	10/22/2026	10/27/2026					X	X						
5	Run initial experiments	All	0	10/22/2026	10/29/2026					X	X						
5	Prepare progress report-Internal Deadline	All	0	10/21/2026	10/30/2026					X	X						
6	Progress report revisions / submission buffer	All	0	10/31/2026	11/6/2026						X	X					
7	Integrate algorithm + visualization	Michael	0	11/7/2026	11/13/2026							X	X				
7	Improve model and catalog standardization	Ugur, Rajesh	0	11/7/2026	11/20/2026							X	X	X			
8	Run final experiments and comparisons	Ugur, Geana	0	11/14/2026	11/24/2026								X	X	X		
8	Refine visualization / usability	Michael	0	11/14/2026	11/24/2026								X	X	X		
8	Write final report + README + clean code -Final Project Internal DeadLine	All	0	11/14/2026	11/27/2026								X	X	X		
10	Final revisions / submission buffer	All	0	11/28/2026	12/4/2026										X	X	
10	Create final poster -Internal Deadline	All	0	11/24/2026	12/1/2026										X	X	
11	Practice individual poster presentations	All	0	12/2/2026	12/8/2026												X

Team responsibilities are tracked in the project Gantt chart; all members have contributed a similar amount of effort.